

GNN–BERT Fusion for Understanding Musical Context

Your Name

Student ID: XXXXXXXX

CSE425: Neural Networks

Department of Computer Science and Engineering

BRAC University

Dhaka, Bangladesh

your.email@g.bracu.ac.bd

Abstract—Understanding musical context requires information from both the acoustic signal and its semantic description. Audio contains timbral, rhythmic, and structural patterns, whereas text can explicitly describe properties such as genre, instrumentation, mood, tempo, and performance characteristics. This project investigates a multimodal framework that combines graph neural networks with Transformer-based text representations for music context understanding.

The work is organized into four tasks. First, a DistilBERT-based multi-label classifier predicts musical aspect tags from MusicCaps captions. Second, FMA-medium tracks are represented as segment graphs and classified using GraphSAGE, with a convolutional neural network on log-mel spectrograms used as an audio baseline. Third, GraphSAGE and DistilBERT representations are combined using early concatenation and cross-attention for multi-context prediction. Finally, a contrastive dual encoder aligns MusicCaps audio graphs and captions in a shared embedding space.

The text classifier achieves a test Macro-F1 of 0.444, Micro-F1 of 0.511, and AUC-PR of 0.453. For FMA genre classification, GraphSAGE reaches 0.307 Macro-F1 and 0.531 Micro-F1, while the CNN baseline reaches 0.311 and 0.614. Early-concatenation fusion performs best in the multimodal experiment, reaching 0.202 Macro-F1, 0.246 Micro-F1, and 0.287 AUC-PR. In cross-modal retrieval, Audio-to-Caption R@10 reaches 3.38% and Caption-to-Audio R@10 reaches 3.70%. These results show that graph-structured audio information and textual context provide complementary signals, while more complex fusion does not necessarily outperform simple feature concatenation when paired training data are limited.

Index Terms—graph neural networks, BERT, GraphSAGE, multimodal learning, music information retrieval, contrastive learning

I. INTRODUCTION

Music conveys several forms of context simultaneously, including genre, instrumentation, rhythm, tempo, mood, production characteristics, and performance style. These properties are not equally represented by a single input modality. Acoustic representations provide direct information about the signal, whereas textual descriptions can express high-level semantic concepts explicitly.

Music information retrieval research commonly uses spectral or time–frequency representations for classification and related tasks. The Free Music Archive (FMA) was introduced as an open dataset containing audio, genre annotations, tags, metadata, and predefined splits for music-analysis research [1].

Although conventional spectrogram-based models are effective at learning local acoustic patterns, they do not explicitly represent relations between distant portions of a track.

Graphs provide a natural alternative when relationships between individual segments are important. GraphSAGE learns node representations by aggregating information from local neighborhoods and supports inductive inference on unseen graphs [2]. In this project, short audio segments are treated as graph nodes and connected through temporal adjacency and acoustic similarity.

Text provides a complementary semantic modality. BERT demonstrated the effectiveness of deep bidirectional Transformer representations for contextual language understanding [3]. DistilBERT reduces the size and computational requirements of BERT through knowledge distillation while retaining much of its language-modeling capability [4]. DistilBERT is therefore used as the text encoder in this project.

MusicCaps provides a direct connection between music and natural language. It was released with MusicLM and contains approximately 5.5K music clips paired with detailed expert-written captions [5]. Related audio–language systems such as CLAP have shown that contrastive learning can map audio and text into a shared representation space [6]. Contrastive predictive coding provides a broader foundation for learning representations by distinguishing positive pairs from negative examples [7].

Motivated by these ideas, this project investigates four settings: text-only musical aspect prediction, graph-based audio classification, supervised GNN–BERT fusion, and contrastive audio–caption retrieval. The main objective is to determine whether structural audio information and semantic text representations provide complementary information for musical context understanding.

II. RELATED WORK

A. Graph Representation Learning

Graph neural networks learn representations by repeatedly combining a node’s features with information from neighboring nodes. GraphSAGE learns an aggregation function rather than a separate embedding for every training node, allowing the model to generalize inductively to unseen graph instances [2].

This property is appropriate for the present problem because every music track forms a separate graph.

In this work, nodes represent short audio segments rather than artists, albums, or songs as abstract entities. The graph therefore provides a structural representation of the recording itself.

B. Contextual Language Models

BERT introduced bidirectional Transformer pretraining in which each token is represented using both its left and right context [3]. DistilBERT later showed that knowledge distillation can produce a smaller and faster Transformer while preserving much of BERT’s performance [4]. This project uses DistilBERT to encode MusicCaps captions and FMA text metadata.

C. Music–Language Learning

MusicCaps contains paired music and natural-language descriptions designed for music-text research [5]. CLAP demonstrates a related approach in the broader audio domain by training separate audio and text encoders using a contrastive objective [6]. The fourth task of this project follows the same general dual-encoder principle but replaces a conventional audio encoder with GraphSAGE over segment graphs.

The contrastive objective is also related to the InfoNCE formulation used in contrastive predictive coding, where positive pairs are distinguished from negative examples within a batch [7].

III. DATASETS AND PREPROCESSING

A. FMA-medium

The Free Music Archive provides audio recordings together with genres, tags, artist metadata, and predefined data partitions [1]. This project uses FMA-medium, containing 25,000 tracks from 5,545 artists and 16 top-level genres.

The official FMA partitions are retained rather than constructing a new random split. The preprocessing pipeline additionally checks for track and artist overlap across partitions. Fifteen corrupted or undecodable tracks identified during preprocessing are excluded.

Only 4,410 FMA-medium tracks contain usable track-level tags. Consequently, Task 3 uses a smaller multimodal subset containing 3,355 training examples, 523 validation examples, and 526 test examples.

B. MusicCaps

MusicCaps contains 5,521 captioned music clips with detailed natural-language descriptions [5]. It is used for Task 1 text classification and Task 4 audio–text retrieval.

Task 1 requires only captions and their musical aspect annotations. Task 4 also requires the underlying audio clips. Because MusicCaps provides YouTube identifiers and timestamps rather than redistributable audio files, the clips were retrieved during preprocessing. Of the 5,521 references, 4,814 clips were successfully available.

The final Task 4 dataset contains 2,070 training pairs, 230 validation pairs, and 2,514 test pairs. The test partition

follows the `is_audioset_eval` indicator provided with MusicCaps.

Table I summarizes the two datasets and their roles in the project. It also highlights the reduction in available samples for Task 3 and the audio-based MusicCaps experiment.

IV. AUDIO GRAPH CONSTRUCTION

A. Audio Segmentation

All audio is converted to mono, resampled to 22,050 Hz, and peak-normalized. Each recording is divided into non-overlapping five-second segments. A final segment shorter than five seconds is zero-padded rather than discarded.

B. Node Features

For each segment, 20 mel-frequency cepstral coefficients (MFCCs) are extracted. The mean and standard deviation of every coefficient are retained, producing 40 MFCC features. A 12-dimensional chroma representation is processed in the same way, contributing another 24 features. Each graph node is therefore represented by a 64-dimensional feature vector.

For a track containing N segments, the node-feature matrix is $X \in \mathbb{R}^{N \times 64}$, where each row corresponds to one audio segment.

C. Graph Edges

The graph combines local temporal structure with non-local acoustic similarity. Consecutive segments are connected through bidirectional temporal edges.

Before acoustic similarity is calculated, every feature dimension is standardized within the current track:

$$\tilde{x}_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}. \quad (1)$$

Cosine similarity between two standardized segment vectors is then

$$s_{ik} = \frac{\tilde{x}_i^\top \tilde{x}_k}{\|\tilde{x}_i\|_2 \|\tilde{x}_k\|_2}. \quad (2)$$

A similarity edge is added when $s_{ik} > 0.3$. Adjacent segment pairs are excluded from this step because they are already connected by temporal edges. This avoids duplicate graph relations between neighboring segments.

V. MODELS

A. Task 1: DistilBERT Tag Classification

Task 1 evaluates how much musical context can be inferred from text alone. MusicCaps captions are used as input, while the target is a multi-hot vector over the 50 most frequent musical aspects in the training split.

Given a token sequence X_{text} , DistilBERT produces contextual token representations

$$H_{\text{text}} = \text{DistilBERT}(X_{\text{text}}). \quad (3)$$

The representation of the first token is used as the sequence-level text embedding:

TABLE I
DATASETS USED IN THE PROJECT.

Dataset	Use	Size / Split
FMA-medium	Tasks 2–3	25,000 tracks from 5,545 artists across 16 genres. Task 3 uses 3,355 training, 523 validation, and 526 test samples.
MusicCaps	Tasks 1, 4	5,521 captioned clips, of which 4,814 audio clips were successfully available. Task 4 uses 2,070 training, 230 validation, and 2,514 test pairs.

$$t = H_{\text{text}}[0]. \quad (4)$$

A linear classification head produces the output logits:

$$o = W_t t + b_t. \quad (5)$$

Independent sigmoid outputs are optimized using binary cross-entropy:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{K} \sum_{k=1}^K [y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)]. \quad (6)$$

The final Task 1 experiment uses `distilbert-base-uncased` [4], a maximum sequence length of 128, and the top two Transformer layers are unfrozen during training.

B. Task 2: GraphSAGE Genre Classification

Task 2 evaluates whether the segment graphs preserve useful genre information. GraphSAGE updates node representations by aggregating information from each node’s neighborhood [2]:

$$h_i^{(\ell+1)} = \phi \left(W^{(\ell)} \left[h_i^{(\ell)} \parallel \text{AGG} \left(\{ h_j^{(\ell)} : j \in \mathcal{N}(i) \} \right) \right] \right). \quad (7)$$

The implemented encoder contains three GraphSAGE layers with hidden dimension 128 and dropout 0.3.

After message passing, global mean pooling converts the node representations into a single graph-level vector:

$$g = \frac{1}{|V|} \sum_{i \in V} h_i^{(L)}. \quad (8)$$

A linear classifier maps g to logits over the 16 FMA genres. Since `genre_top` assigns one mutually exclusive genre to each track, Task 2 is trained using multi-class cross-entropy rather than independent binary outputs.

C. CNN Audio Baseline

A convolutional neural network provides a conventional audio baseline. The CNN receives a 128-bin log-mel spectrogram cropped or zero-padded to a fixed 30-second duration.

This comparison is important because the CNN has access to a richer time–frequency representation, while the GNN uses compact MFCC/chroma summary features but explicitly represents relations between audio segments.

D. Task 3: GNN–BERT Fusion

Task 3 combines the graph representation of an FMA track with text from the corresponding artist biography. The output contains 36 labels: the 16 FMA genres and the 20 most frequent contextual tags selected from the training partition.

Four variants are evaluated: GNN-only, BERT-only, early concatenation, and cross-attention.

1) *GNN-Only*: The GNN-only model uses the graph representation directly:

$$\hat{y} = \sigma(W_g g + b_g). \quad (9)$$

2) *BERT-Only*: The BERT-only model uses only the sequence-level text representation:

$$\hat{y} = \sigma(W_t t + b_t). \quad (10)$$

3) *Early Concatenation*: Early fusion concatenates the global graph and text representations:

$$z_{\text{early}} = [g \parallel t]. \quad (11)$$

The fused representation is then passed to the multi-label classifier:

$$\hat{y} = \sigma(W_f z_{\text{early}} + b_f). \quad (12)$$

4) *Cross-Attention*: For cross-attention, the graph representation is projected into a query:

$$Q = g W_Q. \quad (13)$$

The sequence of DistilBERT token representations is projected into keys:

$$K = H_{\text{text}} W_K. \quad (14)$$

The graph-to-text attention weights are

$$A = \text{softmax} \left(\frac{Q K^\top}{\sqrt{d_t}} \right). \quad (15)$$

The attention-weighted textual context is

$$c = A H_{\text{text}}. \quad (16)$$

The final multimodal representation is

$$z_{\text{attn}} = [g \parallel c]. \quad (17)$$

All four Task 3 variants use the same 36-label target and binary cross-entropy objective, allowing a direct comparison between the fusion strategies.

E. Task 4: Contrastive Audio–Text Alignment

Task 4 learns a common representation space between MusicCaps audio graphs and captions. This follows the general dual-encoder approach used in contrastive audio–language learning [6].

The graph representation is projected into the shared embedding space:

$$z_i^{(g)} = W_g g_i + b_g. \quad (18)$$

The caption representation is projected independently:

$$z_i^{(t)} = W_t t_i + b_t. \quad (19)$$

Both embeddings are L2-normalized:

$$\bar{z} = \frac{z}{\|z\|_2}. \quad (20)$$

For a batch containing N paired audio–caption examples, similarity between audio graph i and caption j is

$$S_{ij} = \frac{\bar{z}_i^{(g)\top} \bar{z}_j^{(t)}}{\tau}, \quad (21)$$

where the temperature is $\tau = 0.07$.

The graph-to-text InfoNCE loss is

$$\mathcal{L}_{g \rightarrow t} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(S_{ii})}{\sum_{j=1}^N \exp(S_{ij})}. \quad (22)$$

The text-to-graph loss is defined analogously. The final objective averages both retrieval directions:

$$\mathcal{L}_{\text{NCE}} = \frac{1}{2} (\mathcal{L}_{g \rightarrow t} + \mathcal{L}_{t \rightarrow g}). \quad (23)$$

This objective belongs to the broader family of contrastive representation learning methods associated with InfoNCE [7]. Retrieval is evaluated using Recall@1, Recall@5, and Recall@10 in both directions.

VI. EXPERIMENTAL SETUP

All experiments are implemented using PyTorch and PyTorch Geometric. A random seed of 42 is used for reproducibility. The general training configuration uses a maximum of 30 epochs, batch size 32, and weight decay 0.01.

Task 1 and Task 3 use `distilbert-base-uncased` with a maximum sequence length of 128. In their final runs, the top two Transformer layers are unfrozen and the learning rate is 2×10^{-5} .

The GraphSAGE encoder uses three layers, hidden dimension 128, and dropout 0.3. The final Task 2 run uses a learning rate of 3×10^{-4} and early-stopping patience 10.

Task 4 uses a 128-dimensional joint embedding space and a temperature of 0.07. The DistilBERT branch remains frozen in the reported contrastive run.

For multi-label classification, prediction thresholds are tuned using the validation partition only and then fixed before test evaluation. Macro-F1 and Micro-F1 are used as the primary

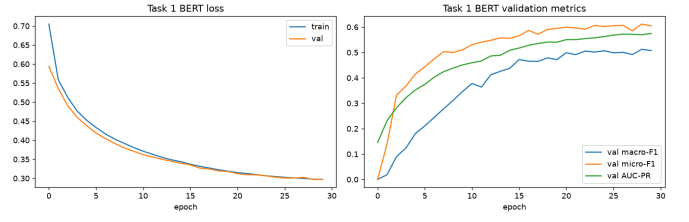


Fig. 1. Training and validation curves for the Task 1 DistilBERT classifier.

classification metrics, while AUC-PR is also reported for the multi-label tasks. Cross-modal retrieval is evaluated using Recall@ K .

VII. RESULTS

Table II summarizes the final classification results for Tasks 1–3. Results from different tasks should not be compared directly because their datasets and target spaces differ. The four Task 3 variants, however, share the same data and labels and therefore form a direct ablation.

A. Task 1: Text-Based Context Prediction

As shown in Table II, the DistilBERT caption classifier achieves 0.444 Macro-F1, 0.511 Micro-F1, and 0.453 AUC-PR. This establishes a strong text-only baseline.

MusicCaps captions frequently describe instrumentation, tempo, vocal characteristics, mood, and recording properties explicitly. The result is therefore consistent with the strong semantic supervision provided by the MusicCaps descriptions [5].

Figure 1 presents the training and validation curves for Task 1. These curves allow the progression of optimization and validation performance to be inspected throughout fine-tuning.

Figure 2 shows qualitative caption examples together with token-level attention visualizations. The visualization helps inspect which portions of a caption receive greater attention for example predictions, although attention values should be interpreted as supplementary analysis rather than a complete causal explanation of the model.

B. Task 2: Graph-Based Genre Classification

Table II shows that both learned audio models substantially outperform the majority baseline. GraphSAGE reaches 0.307 Macro-F1 and 0.531 Micro-F1, demonstrating that the segment graph preserves useful genre information.

The CNN baseline performs better overall, reaching 0.311 Macro-F1 and 0.614 Micro-F1. One likely explanation is the difference in input representation. The GNN compresses each five-second segment into 64 MFCC/chroma statistics, whereas the CNN receives a richer log-mel time–frequency representation. The graph therefore gains explicit relational structure while sacrificing some fine-grained spectral information.

The difference between the two models is small in Macro-F1 but considerably larger in Micro-F1. This suggests that the CNN’s advantage is stronger for more frequently represented classes, which have greater influence on the micro-averaged score.

TABLE II
FINAL TEST RESULTS FOR TASKS 1–3.

Task	Model	Macro-F1	Micro-F1	AUC-PR
1	DistilBERT caption classifier	0.444	0.511	0.453
2	Majority baseline	0.027	0.276	–
2	GraphSAGE segment graph	0.307	0.531	–
2	CNN log-mel baseline	0.311	0.614	–
3	GNN-only	0.152	0.152	0.196
3	BERT-only	0.139	0.146	0.232
3	Early concatenation	0.202	0.246	0.287
3	Cross-attention	0.153	0.193	0.199

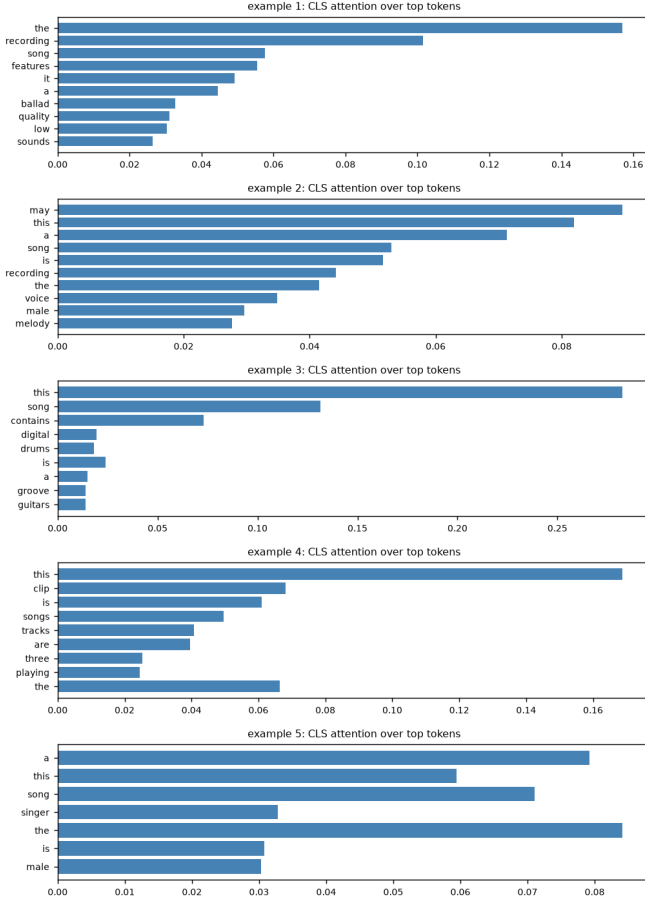


Fig. 2. Example Task 1 predictions with DistilBERT attention visualizations.

C. Task 3: Multimodal Fusion

The Task 3 rows of Table II form the primary multimodal ablation. The GNN-only model achieves 0.152 Macro-F1, while the BERT-only model reaches 0.139. The BERT-only model nevertheless has a higher AUC-PR, indicating that probability ranking and thresholded F1 do not necessarily favor the same model.

Early concatenation is the strongest variant on all reported metrics, reaching 0.202 Macro-F1, 0.246 Micro-F1, and 0.287 AUC-PR. Its Macro-F1 is approximately 33% higher than that

TABLE III
TASK 4 MUSICCAPS CROSS-MODAL RETRIEVAL RESULTS.

Direction	R@1	R@5	R@10
Audio → Caption	0.36%	1.91%	3.38%
Caption → Audio	0.44%	2.31%	3.70%

of the GNN-only model. This supports the central hypothesis that structural audio information and semantic text information are complementary.

Figure 3 shows the validation trajectories of the four Task 3 variants. The figure complements the final test results in Table II by showing how their performance develops during training.

Cross-attention reaches only 0.153 Macro-F1 and therefore does not outperform early concatenation. This should not be interpreted as evidence that attention-based multimodal fusion is generally ineffective. The Task 3 training set contains only 3,355 paired examples, and the text modality is an artist biography rather than a description aligned directly with an individual track. Cross-attention must therefore learn relatively fine-grained relationships from weakly aligned supervision.

Figure 4 provides a qualitative t-SNE projection of the fused Task 3 representations. It is useful for visually inspecting local grouping among representations, but it is not treated as a quantitative measure of model quality. The metrics in Table II remain the primary basis for comparison.

D. Task 4: Cross-Modal Retrieval

Task 4 evaluates whether the contrastive model can retrieve the matching modality without using supervised context labels. The final results are shown in Table III.

As Table III shows, Caption-to-Audio retrieval is slightly stronger than Audio-to-Caption retrieval at all three values of K . Absolute retrieval performance remains low, showing the difficulty of identifying one paired example among 2,514 candidates.

Random R@10 for this candidate set is approximately 0.40%. The observed R@10 values of 3.38% and 3.70% are therefore roughly eight to nine times higher than random retrieval. The model consequently learns a non-trivial cross-modal signal even though its embedding space is not sufficiently discriminative for high-accuracy retrieval.

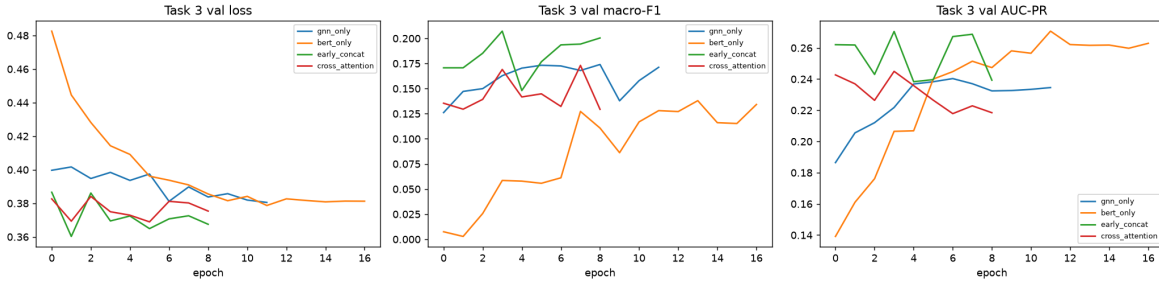


Fig. 3. Validation curves for the Task 3 GNN-only, BERT-only, early-concatenation, and cross-attention variants.

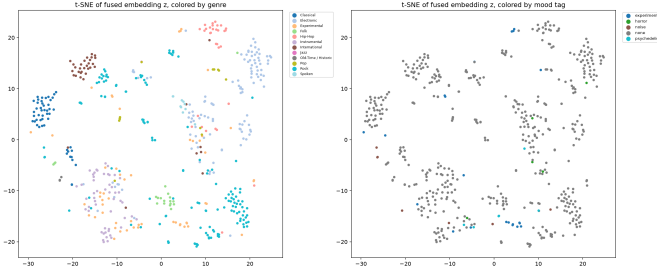


Fig. 4. t-SNE visualization of fused Task 3 representations.

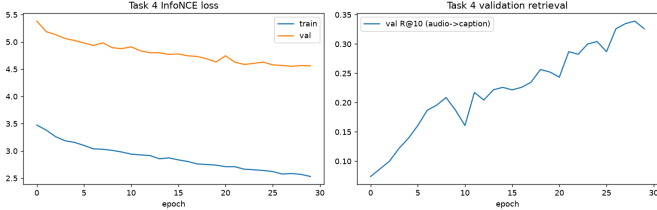


Fig. 5. Training and validation behavior of the Task 4 contrastive dual encoder.

Figure 5 shows the training and validation behavior of the contrastive dual encoder. Together with Table III, the figure provides a view of both the optimization process and the final retrieval performance.

VIII. DISCUSSION

The experiments produce three main observations.

First, text is highly informative when it directly describes a recording. Task 1 achieves substantially stronger text-only performance than Task 3. This difference is consistent with the nature of the datasets: MusicCaps was specifically created around detailed music descriptions [5], whereas an FMA artist biography describes an artist more generally and is not guaranteed to describe an individual track.

Second, graph-based audio modeling is viable but strongly dependent on the quality of the node representation. The GraphSAGE classifier substantially outperforms the majority baseline, showing that segment features and graph relations preserve meaningful genre information. However, the CNN’s stronger Micro-F1 suggests that the compact MFCC/chroma representation discards acoustic detail retained by the log-mel

spectrogram. A stronger future graph model could therefore use learned audio representations as node features before message passing.

Third, the Task 3 ablation provides direct evidence that the two modalities are complementary. Neither GNN-only nor BERT-only reaches the performance of early concatenation. At the same time, the weaker cross-attention result demonstrates that greater architectural complexity does not automatically produce better multimodal learning. Dataset size, alignment quality, and optimization difficulty also matter.

The contrastive experiment leads to a related conclusion. Systems such as CLAP show that sufficiently large paired audio-text datasets can support useful joint representations [6]. The present experiment contains only 2,070 paired training examples and uses a compact graph-based acoustic representation. Its above-random retrieval performance is therefore encouraging, but more paired data and stronger acoustic representations would likely be required for high-accuracy retrieval.

IX. LIMITATIONS

The project has several limitations. First, the usable multimodal datasets are relatively small. Although FMA-medium contains 25,000 tracks, only 4,410 have usable track-level tags, leaving 3,355 training examples for Task 3.

Second, artist biography text is only indirectly aligned with the corresponding recording. It may contain useful information about genre, style, history, or influences but does not necessarily describe the acoustic content of a specific track.

Third, the graph branch uses fixed MFCC/chroma summary statistics. These features make graph construction computationally manageable but discard fine-grained temporal and spectral information.

Fourth, only 4,814 of the 5,521 MusicCaps audio references could be retrieved because some of the underlying source videos were unavailable.

The optional DEAM valence/arousal extension described in the original project specification was not included in the final experiments. Task 3 instead uses 16 genre labels together with 20 data-driven contextual music tags. These results should therefore not be interpreted as a dedicated emotion-recognition benchmark.

Finally, the reported values correspond to the final project runs rather than an extensive hyperparameter search. The

comparisons support conclusions within this implementation but should not be interpreted as universal rankings of the tested architectures.

X. REPRODUCIBILITY

The implementation is organized as a reproducible PyTorch project. A central `config.yaml` file stores dataset, preprocessing, graph, BERT, GNN, fusion, contrastive, and optimization parameters.

Separate source modules implement audio feature extraction, graph construction, GraphSAGE, the CNN baseline, DistilBERT encoding, multimodal fusion, contrastive learning, training, and evaluation. The repository also stores dataset split information, experiment metrics, logs, plots, example graph objects, and a demonstration notebook.

Automated tests cover dataset loading, split integrity, graph construction, audio feature extraction, tensor shapes, and leakage-related assumptions. A continuous-integration workflow runs the test suite on repository updates.

Raw dataset audio and trained checkpoints are not committed because of data distribution requirements and file size. The repository instead contains the configuration and scripts required to reproduce preprocessing and training.

XI. CONCLUSION

This project investigated multimodal musical context understanding by combining graph neural networks with contextual language models.

Audio recordings were divided into five-second segments and represented as graphs containing temporal and acoustic-similarity edges. GraphSAGE learned structural representations from these graphs, while DistilBERT encoded textual context.

The experiments show that both modalities contain useful but different information. The Task 1 text classifier successfully predicts musical aspects from MusicCaps captions, while Task 2 demonstrates that GraphSAGE learns genre-relevant information from segment graphs. Most importantly, the Task 3 early-concatenation model outperforms both individual modalities, supporting the use of combined structural and semantic representations.

The cross-attention model does not outperform simple concatenation, highlighting the importance of data quantity and cross-modal alignment rather than architectural complexity alone. Contrastive training also produces audio-caption retrieval substantially above random chance, although absolute retrieval performance remains limited.

Future work could replace fixed MFCC/chroma node features with learned or pretrained audio embeddings, use text directly aligned with individual tracks, increase the amount of paired training data, and investigate stronger graph-text alignment strategies.

REFERENCES

[1] M. Defferrard, K. Benzi, P. Vandergheynst, and X. Bresson, "FMA: A dataset for music analysis," in *Proceedings of the 18th International Society for Music Information Retrieval Conference*, 2017, pp. 316–323.

[2] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[3] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, 2019, pp. 4171–4186.

[4] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter," *arXiv preprint arXiv:1910.01108*, 2019.

[5] A. Agostinelli, T. I. Denk, Z. Borsos, J. Engel, M. Verzett, A. Caillon, Q. Huang, A. Jansen, A. Roberts, M. Tagliasacchi, M. Sharifi, N. Zeghidour, and C. Frank, "MusicLM: Generating music from text," *arXiv preprint arXiv:2301.11325*, 2023.

[6] B. Elizalde, S. Deshmukh, M. A. Ismail, and H. Wang, "CLAP: Learning audio concepts from natural language supervision," in *2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.

[7] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," *arXiv preprint arXiv:1807.03748*, 2018.